

Submission CALC1-MID-SUB02

Question CALC1-MID-Q01

work / setup:

1. $f'(x) = 2x(2x-1) + (x^2+4)^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 10 plus 1

Question CALC1-MID-Q02

work / setup:

1. $\int 2x \, dx = (2/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 9 plus 1

Question CALC1-MID-Q03

work / setup:

1. $2w+2l=22 \Rightarrow l=11-w$

2. $A(w)=w(11-w)$; at $w=8$, $A=24$

crossed-out arithmetic: [smudged]

FINAL (maybe): $A(w)=w(11-w)$; $A(8)=24$; check $A'(w)=0$ plus 1

Question CALC1-MID-Q04

work / setup:

1. $x^2-16=(x-4)(x+4)$

2. cancel $x-a$ only after identifying $x \neq a$

crossed-out arithmetic: [smudged]

FINAL (maybe): 8 plus 1

Question CALC1-MID-Q05

work / setup:

1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).

crossed-out arithmetic: [smudged]

FINAL (maybe): critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum plus 1

Question CALC1-MID-Q06

work / setup:

1. $f'(x) = 2x(2x-1) + (x^2+2)^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 14 plus 1

Question CALC1-MID-Q07

work / setup:

1. $\int 2x \, dx = (2/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 16 plus 1

Question CALC1-MID-Q08

work / setup:

1. $2w+2l=24 \Rightarrow l=12-w$

2. $A(w)=w(12-w)$; at $w=9$, $A=27$

crossed-out arithmetic: [smudged]

FINAL (maybe): $A(w)=w(12-w)$; $A(9)=27$; check $A'(w)=0$ plus 1